Anqi Tan

Optical & Hardware Systems Engineer · Portable Biomedical Instruments

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Ph.D. candidate who designs portable optical instruments end to end: optics, mechanical design, embedded hardware, and the Android and analysis software that make them usable.

EDUCATION

Ph.D. in Bioengineering, University of Illinois Urbana-Champaign Aug 2022 – Expected Dec 2027

Advisor: Prof. Brian T. Cunningham · Nanosensors Group

B.S. in Mechanical Engineering (dual degree), UIUC & Zhejiang University Aug 2018 – Jun 2022

EXPERIENCE

Graduate Research Assistant, Nanosensors Group, UIUC Sep 2022 – Present

- Built a **$87 pocket-size NIR fluorescence reader** end to end: 750 nm excitation and custom optical filtering, Raspberry Pi imaging hardware, and an **Android control app (Kotlin)** with automated T/C analysis in **20 s per strip**; agrees with a lab ImageQuant scanner (**R^2 = 0.99**, **3.2% CV**); first-author manuscript in preparation.
- Wrote the reader's analysis pipeline (**automated test/control-line detection, background correction, 4PL calibration**) that converts strip images into cortisol concentrations.
- Lead developer of the portable fluorescence reader for a **$4.87M DoD (PRMRP)** project on point-of-care sepsis/ARDS phenotyping with Stanford, WashU, and UCSF.
- Developed a plasmonic-fluor lateral flow assay for salivary cortisol (**≤0.12 ng/mL** LOD; **153 adults** sampled); **oral presentation, APHA 2026**.

Undergraduate Research Leader, ZJU-UIUC Institute 2019 – 2021

- Led a national-level research project designing a re-localizable remote-center-of-motion mechanism for minimally invasive surgery (**first-author paper, IEEE ICARM 2021**).

SELECTED PROJECTS

Mimicry Stage Lighting Control System (Senior Design, mechanical lead) Feb 2022 – May 2022

- Designed a two-axis light manipulator (MATLAB kinematics, Euler–Lagrange torque analysis) and a 3D-printed/laser-cut portable structure (9.3 kg, 4.5 h battery); the light follows a handheld ESP32 + IMU controller in real time.

Intelligent Cell Culture Platform (China "Internet+" Competition) — National Silver Medal 2021

- Product lead for an automated, modular manipulator platform for standardized cell production.

PUBLICATIONS

First author: IEEE ICARM 2021 paper · APHA 2026 oral presentation · pocket fluorescence reader manuscript (in preparation). **Co-author:** 4 journal papers (*Chemical Reviews*, *Lab on a Chip*, *ACS Nano Medicine*, *Biosensors*) and 1 Elsevier book chapter.

HONORS & AWARDS

1st Place, Ashby Prize Computational Science Hackathon, NCSA (2025) · Honorable Mention, Ashby Prize Hackathon (2024) · National Silver Medal, China "Internet+" Innovation & Entrepreneurship Competition (2021)

TEACHING

TA, BIOE/ECE 415 Biomedical Instrumentation Lab (Fall 2025) · TA, ME 170 Computer-Aided Design (3 semesters)

SKILLS

Optics & Hardware: fluorescence imaging optics (NIR excitation, filter selection), camera modules, Raspberry Pi, ESP32, Arduino, LabVIEW, IMU sensors

Engineering: SolidWorks, Creo, Fusion 360, Blender, 3D printing, laser cutting, optical/instrument design, COMSOL, FEA, design for assembly

Software: Python, MATLAB, Android app development (Kotlin), image processing, curve fitting (4PL)

Languages: English (fluent), Chinese (native)